

1. Introduction

Fundamentals of Engineering Materials

Condensed study notes · compiled 2026-09-19

1 Materials Science and Engineering

Definition: Materials Science

Investigates relationships between structures and properties of materials, and designs/develops new materials. Source: 2.PNG

Definition: Materials Engineering

Creates products from existing materials and develops materials processing techniques. Source: 2.PNG

Definition: Core terms

- **Composition:** chemical make-up of a material.
- **Structure:** arrangement of atoms in a material.
- **Processing:** ways to shape materials into useful components.

Source: 2.PNG

The materials chain

Composition → Structure → Processing → Properties → Performance

Source: 2.PNG

Definition: Materials knowledge spectrum

Materials science provides *basic* knowledge; materials engineering provides *applied* knowledge. Together they produce the resultant knowledge of structure, properties, processing and performance that engineers use to convert materials into the products needed by society. Source: 6.PNG

Example: Structure, not composition, sets properties

A disk of aluminum oxide in three forms (single crystal, polycrystalline, and polycrystalline with many pores) has different optical properties in each form despite identical composition. Source: 3.PNG

2 Why Materials Are Important

Definition: Why it matters

Materials drive advancements in society. Engineers must understand materials in order to select appropriate materials and processing techniques for specific applications. Source: 4.PNG, 5.PNG

Definition: Historical materials ages

- Stone Age, Bronze Age, Iron Age mark society's advancement.
- Today's age is posed as an open question. The candidates offered are the Silicon (Electronic Materials) Age, the Nano-materials Age, and the Polymer Age.

Source: 4.PNG

Definition: Engineer's requirement

Products, devices and components are all made of materials. Selecting appropriate materials and processing requires knowledge of material properties and of structure–property relationships.

Source: 5.PNG

3 Types of Materials

Definition: Three broad classes

Metals and alloys; ceramics and glasses; polymers and plastics. Source: 7.PNG

3.1 Metals and Alloys

Definition: Alloy

A metal that contains additions of one or more metals or non-metals. Source: 8.PNG

Definition: Metals: structure and properties

- Pure metallic elements or combinations of them; a large number of de-localized electrons, and therefore able to conduct electricity. Source: 7.PNG
- Properties: good thermal and electrical conductivity, high strength, high stiffness, ductility. Source: 8.PNG
- Steel is an alloy of iron and a small content of carbon. Brass is an alloy of copper and zinc. Source: 8.PNG
- Examples: aluminum, magnesium, zinc, iron, titanium, beryllium-copper, nickel, copper-nickel, niobium. Source: 8.PNG

3.2 Ceramics and Glasses

Definition: Ceramics

Compounds of metallic and non-metallic elements: oxides, carbides, nitrides, sulfides; generally insulating and refractory. Source: 7.PNG Inorganic non-metallic materials, crystalline (polycrystalline) in structure; examples include sand and rocks. Properties: insulating (non-conductive), high hardness, high strength and melting point, very brittle. Source: 9.PNG

Definition: Glasses

Amorphous materials with structure-dependent properties, which can be processed to yield an improvement in properties. Source: 9.PNG

3.3 Polymers and Plastics

Definition: Polymer

“Poly” means many, “mer” means unit. Polymers are large molecular structures formed from monomers through polymerization. Source: 10.PNG

Definition: Polymers: composition and properties

- Chemical composition including carbon, hydrogen and other non-metals; large molecular structure; flexible, lightweight, easily processed. Source: 7.PNG
 - Synthetic: polyethylene (PE), polymethyl methacrylate (PMMA), polycarbonate. Natural: cellulose, glucan. Source: 10.PNG
- **Practice.** Classify as metal, ceramic or polymer: brass, MgO, plexiglass, polychloroprene, B₄C, cast iron. The source provides no answer key. Source: 14.PNG

4 Other Materials

Definition: Composite

A combination of two or more different materials, combining the properties of the materials it is formed from. Source: 11.PNG, 12.PNG

Definition: Advanced materials

Semiconductors, biomaterials, smart materials, and nanoengineered materials. Source: 11.PNG

Definition: Semiconductors

Conductivity in between ceramics and metals. Used in computers, electronics and micro-electrical-mechanical systems (MEMS). Examples: silicon, germanium, gallium arsenide. Source: 13.PNG

5 Comparing Material Properties

Definition: Material property chart

Compares classes of materials (metals, ceramics, polymers, composites) against measurable properties such as density, elastic modulus and tensile strength, often on logarithmic scales. Source: 15.PNG

Ranking by class

Property	Highest	Lowest
Density	metals	polymers
Elastic modulus	metals, ceramics	polymers
Tensile strength	composites, metals, ceramics	none

Composites and woods occupy a wide middle range for density and stiffness. For tensile strength every class spans a range, so no class is lowest outright. Source: 15.PNG, 16.PNG, 17.PNG

Definition: Ashby map (two-property optimization)

Plots stiffness (elastic/Young's modulus, GPa) against density (kg/m^3) on logarithmic axes, grouping materials into regions for technical ceramics, non-technical ceramics, composites, metals, natural materials, polymers, foams and elastomers. Used to compare and select materials on two properties simultaneously. Source: 18.PNG

6 Materials Selection Process

Method: Selecting a material and process

1. For a specific application, determine the required properties: mechanical, electrical, thermal, magnetic, optical, deteriorative.
2. From the list of required properties, identify candidate material(s).
3. For the best candidate material(s), specify processing technique(s), to provide the required set of properties and to produce a component of the desired shape and size. Example techniques: casting, mechanical forming, welding, heat treating.

Source: 19.PNG